Raaha Sellamuthu

I aspire to become a cardiothoracic surgeon-scientist, administering life-saving procedures and researching innovative tissue engineered solutions.

(321) 344-3775 raaha.vk@gmail.com raahavk.com

EDUCATION

West Shore Jr/Sr High School, Melbourne FL: (2023-2027)

- AP Capstone Program (18 APs): Biology, Chemistry, Physics I, Calculus AB, Microeconomics, Lang. and Comp., Capstone Seminar, Comp. Sci. Principles, World History, Precalculus, Spanish Lang. and Culture, US History, Literature, Psychology, Calculus BC, Physics 2, Statistics, Capstone Research
- Mu Alpha Theta, National Honor Society, Spanish Honor Society, Tri-M Music Honor Society
- Unweighted/Weighted GPA: 4.0/4.61
- Class Rank: 1/136
- 1550 SAT/35 ACT

CERTIFICATIONS

- **STOP THE BLEED - American College of Surgeons: (June 2025)**
  - Learned how to recognize and stop life-threatening bleeding by applying direct pressure, packing wounds, and using a tourniquet.
- **Brain Health Peer Educator - Hilarity for Charity (HFC): (November 2025)**
  - Passed knowledge and presentation evaluations on the five main brain health habits. Used skills learned to educate students and the community via outreach events.

HEALTHCARE ACTIVITIES

Vascular Tissue Engineering Research - *Research Assistant* (September 2024-Present)

- Worked in Bashur Vascular Tissue Engineering Lab at Florida Tech (35 wks/yr & 6-8 hrs/wk)
- Project 1 (2025): developed pH-responsive particles that can deliver pro-angiogenic substances such as carbon monoxide in acidosis conditions in the context of ischemia developing in the graft
  - Conducted laboratory experiments, including synthesizing particles and operating lab equipment, and data analysis
- Project 2 (2026)—**pending publication**: investigated the spatio-temporal availability of transforming growth factor-β3 within a bioprinted gradient collagen sleeve to promote chondrogenic differentiation in the context of ACL reconstructions
  - Cultured cells, quantified 100+ images, made graphs, curated images for publication figures, and ran statistical tests.
  - Oral Presentation at TERMIS-AM Annual Conference (November 2026)

Health Occupations Students of America–Future Health Professionals - *Treasurer* (August 2023-Present)

- Raised $475 towards competition fees via a “Pie a Teacher” event, Spring Festival Water Balloon Dodgeball Booth, and Crumbl Cookie fundraiser (32 wks/yr & 1 hr/wk)
- 2026 International Competitor for Clinical Specialty; two-time state competitor for Dental Terminology event and Clinical Specialty presentation on geriatrician
- Learned CPR and how to perform an EKG test

Shadowed Geriatrician, Pediatrician, and Rheumatologist (December 2025-Present)

- Observed differences in routine check-ups, treatments, and priorities of care (50 hours total)
- Witnessed clinical workflow, multidisciplinary team communication, and complex cases

COMMUNITY SERVICE

Mindcare Medics - *President/Founder* (April 2025-Present)

- Centered around educating the community about brain health, breaking the stigma against Alzheimer's and ageism, and supporting local seniors (40 wks/yr & 2-4 hrs/wk)
- Hosted outreach events for 250+ Juniors and Freshmen from West Shore with two guest speakers and hands-on activities that explored the effects of aging and differentiated them from Alzheimer's symptoms
- Partnered with the Health First Memory Disorder Clinic and the One Senior Place to host 18 intergenerational craft/tech classes connecting high school students with ~180 seniors
- Fundraised $310 via a Holiday Bake Sale and Spring Festival Trivia Booth; $150 of earnings were donated to Meals on Wheels program at Aging Matters
- Directed community health fair with 10+ student-led booths, 5+ local organizations, 2 food vendors, live music, and prize giveaways to raise awareness about cognitive health and fundraise for a local aging nonprofit

Baba Senior Outreach - *Intergenerational Outreach Committee President* (February 2026 - Present)

- Connected with Helping Seniors of Brevard to host technology workshops and bring a youth perspective to digital literacy (40 wks/yr & 3 hours/week)
    - 2 workshops on AI in daily life and cellphone tips (~50 seniors)
    - Featured in their radio broadcast and article
- Expanded Mindcare Medics to Melbourne High School (2026)
- To reach fundraising goal of $50,000, I helped raise $15,000 towards expanding grocery and housekeeping services to 200+ seniors
- Created video highlighting experiences and hardships of seniors in the community that was featured at Baba Walk and Show event

Health First Holmes Regional Medical Center - *Music MDs Training Officer/Musician & Emergency Room Volunteer* (March 2024 - Present)

- **MusicMDs**: Perform classical music for patients/staff in a local hospital to reduce stress and promote patient well-being/normalcy; train new volunteers on proper protocol for patient interactions
- **ER**: Restock supplies in patient rooms and serve as an advocate for patient needs, often communicating with physicians and nurses
- 40 wks/yr & 4-7 hours/week (234 hours total)

SUMMER ENRICHMENT

FSU SSTRIDE Summer Institute Participant: (June 2025)
Further explored my passion for the medical field by shadowing an open heart surgery, learning clinical skills, performing a heart dissection, and completing a team research project.

University of Florida's Student Science Training Program (2026)
Admitted as top 10-11% of applicants to attend seven-week research program to conduct research in a field of interest, write a research paper, give two research talks, and present a research poster.

LEADERSHIP EXPERIENCE

Future Problem Solvers (FPS) - *Co-President/Student Coach/Competitor*
(August 2021-Present)

- Two-time International Individual competitor (2024/2025)—utilized 6-step problem-solving process to resolve issues in socio-economic future scene scenarios within two hours
- 2026 International Conference 3rd Place for a Community Problem Solving Project spreading awareness about nitrogen pollution in the Indian River Lagoon
    - Designed curriculum (250+ elementary students), managed social media, coordinated with school administrators to organize 6 events, and expanded nonprofit statewide
- Trained/evaluated new students on writing competition booklets via the 6-step process
    - Oversee 10+ student coaches, develop educational material, and lead meetings
- 25 wks/yr & 2-3 hrs/wk

Manav Mandir Temple - *Indiafest Student Choreographer* (January 2025 - Present)

- Choreographed two Bollywood dance routines for elementary school students (15 wks/yr & 4 hrs/wk)
- Taught the dances, ran weekly rehearsals, and prepared students for two on-stage performances

SUMMER ENRICHMENT

HOBY Florida-East Participant (June 2025)

- Nominated by school to attend a 3-day leadership seminar through 3 phases: Personal, Group, and Service.
- Participated in team-building challenges with a group full of peer leaders, listened to inspiring lectures, and led four mini service projects.

INTERESTS

Music - *Indian Carnatic Singer, Violinist, Pianist* (August 2017-Present)

- Learned Indian Carnatic singing for 10 years and performed at multiple religious venues
- 5x All-County and 2x All-State violin (one of the top 32 violins in FL); learned violin for 7 years; concertmaster of school Chamber Orchestra; in Brevard String Academy's Premier Orchestra
    - Ensemble won 1st place at National Orchestra competition (ASTA)
    - Ensemble performed at Steimetz Hall in Orlando before an Orlando Philharmonic concert
    - Top 5 in Indian River Symphonic Concerto competition (April 2026)
- Two-time state honors piano recital performer (top 15 students in state)
- 40 wks/yr & 4-6 hours/week

HONORS

- **National Merit Semifinalist**
- **2026 Gerson Rising Leaders in Aging Services Award Recipient ($250)**
- **1st Place Team - Precalculus Division at District Math Tournament**
- **Golden Seal of Biliteracy in Spanish/1st Place Spanish III at Brevard Spanish Spelling Bee**
- **3rd Place Brevard Science Fair in Cellular and Molecular Biology/Biochemistry category/$84,000 scholarship to FIT based on promised potential (2026)**
- **2x 100 Hours of Community Service reward recipient/5x Honor Roll (4.0 GPA)**
- **Top Student in Civics, Geometry, Algebra II, AP Calc AB, AP Spanish Lang. & Culture**
- **Fluent in Tamil/English & Conversational in Spanish**